

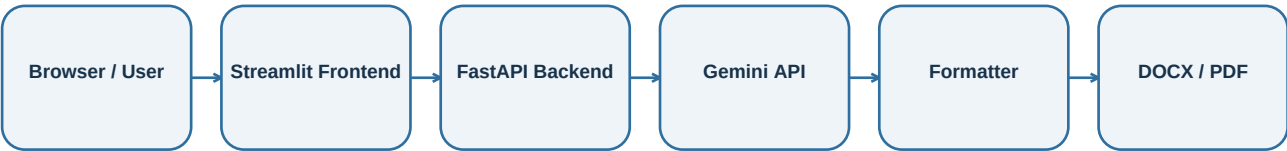
LegalEase: AI-Powered Legal Document Generator

Solution Architecture

Project Phase	3. Project Design Phase
Team	Team 3
Date	02 October 2026
Prepared From	Supplied LegalEase repository + supplied project templates

Runtime architecture

High-level runtime



Layer responsibilities

Layer	Responsibility	Repository evidence
Presentation	Collect inputs, errors, preview and downloads	frontend/app.py + style.css
API	Validate request and expose generation endpoint	legalEaseAPI/routes.py
AI service	Build prompt, add document guidance, retry generation	ai_core/gemini_generator.py
Formatting	Sanitize and create HTML/DOCX/PDF	ai_core/generator.py
Configuration	Load model, API URL, key and logo path	config.py + .env.example
Testing	Verify formatting and API behavior	tests/test_flow.py

Request sequence

User → Streamlit form → POST /generate → FastAPI validation → Gemini prompt → generated text → preview → DOCX/PDF formatter → download.

Source basis: Implemented claims are based on the supplied LegalEase repository. Items without repository evidence are explicitly marked pending, future, or not implemented.